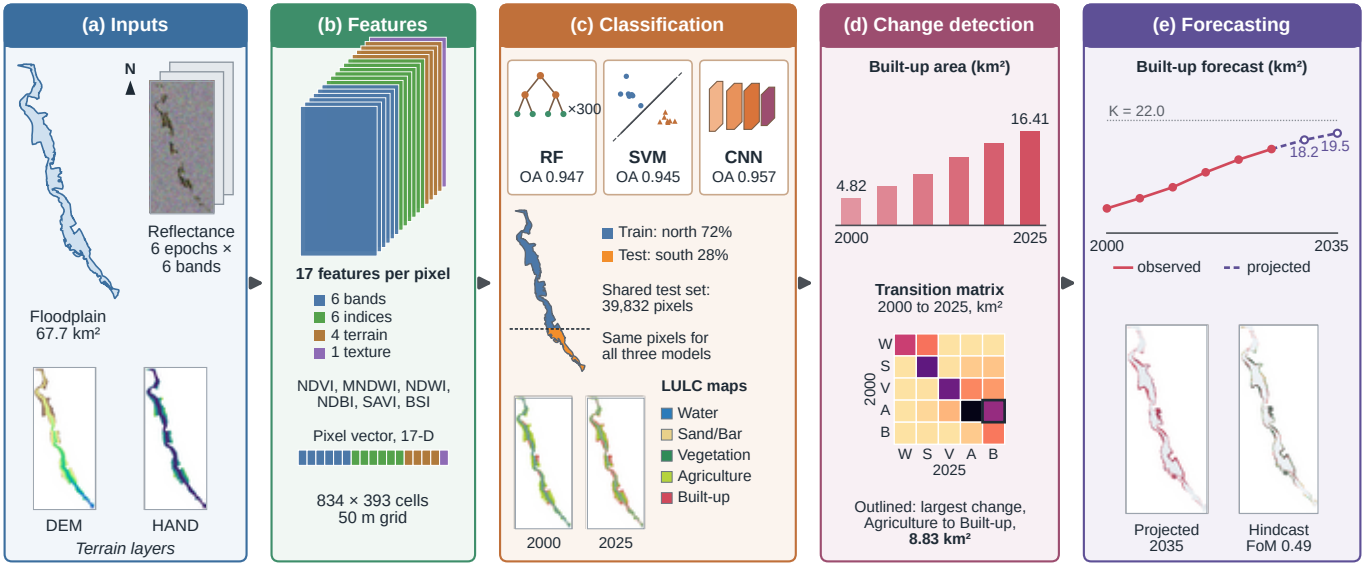

(a) Inputs
N
Reflectance
6 epochs ×
6 bands
Floodplain
67.7 km²
DEM
HAND
Terrain layers
(b) Features
17 features per pixel
6 bands
6 indices
4 terrain
1 texture
NDVI, MNDWI, NDWI,
NDBI, SAVI, BSI
Pixel vector, 17-D
834 × 393 cells
50 m grid
(c) Classification
×300
RF
OA 0.947
SVM
OA 0.945
CNN
OA 0.957
Train: north 72%
Test: south 28%
Shared test set:
39,832 pixels
Same pixels for
all three models
LULC maps
Water
Sand/Bar
Vegetation
Agriculture
Built-up
2000
2025
(d) Change detection
Built-up area (km²)
4.82
16.41
2000
2025
Transition matrix
2000 to 2025, km²
W
S
V
A
B
2000
W S V A B
2025
Outlined: largest change,
Agriculture to Built-up,
8.83 km²
(e) Forecasting
Built-up forecast (km²)
K = 22.0
18.2
19.5
2000
2035
observed
projected
Projected
2035
Hindcast
FoM 0.49